Year 13 Further Maths - Further calculus techniques

Similar questions - Maclaurin series

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Use differentiation to find the Maclaurin series of $\ln(1+\sin x)$ up to and including the term in x^3. **[5]**

(b) Hence estimate $\ln(1+\sin 0.2)$ to 4 decimal places and compare with the calculator value. **[2]**

2 Let $f(x) = \tan^{-1}(2x+1)$.

(a) Find $f(0)$, $f'(0)$ and show that $f''(0) = -2$. **[5]**

(b) Hence find the Maclaurin series of $f(x)$ up to and including the term in x^2. **[2]**

3 Let $f(x) = \sin^{-1} x$.

(a) Find $f''(x)$ and the first two non-zero terms of the Maclaurin series for $f(x)$. **[5]**

(b) Use the first two non-zero terms to approximate $\int_0^{1/2} f(x)\,dx$ to 6 decimal places. **[2]**

(c) By writing $f(x) = \sin^{-1} x \times 1$, find the exact value of the integral. **[3]**

4

(a) Find the Maclaurin series of $(\ln(1+x))^2$ up to and including the term in x^4. **[3]**

(b) The curves $y = (\ln(1+x))^2$ and $y = 2x^3$ meet at the origin and at a point A with $x > 0$. Use your series to find an approximation to the x-coordinate of A to 2 decimal places. **[3]**

5 In this question you must show detailed reasoning.

(a) By using an appropriate Maclaurin series, prove that if $x > 0$ then $e^x > 1+x$. **[2]**

(b) Hence, by a suitable substitution, deduce that $e^t > et$ for $t > 1$. **[1]**

(c) By making a suitable choice of t, determine which is greater, e^4 or 4^e. **[3]**

Question 1

(a) $f = \ln(1+\sin x)$, $f(0) = 0$. $f' = \dfrac{\cos x}{1+\sin x}$, $f'(0) = 1$. $f'' = \dfrac{-\sin x(1+\sin x) - \cos^2 x}{(1+\sin x)^2} = \dfrac{-\sin x - 1}{(1+\sin x)^2} = -\dfrac{1}{1+\sin x}$, so $f''(0) = -1$. $f^{(3)} = \dfrac{\cos x}{(1+\sin x)^2}$, $f^{(3)}(0) = 1$. Series: $x - \dfrac{x^2}{2} + \dfrac{x^3}{6}$

(b) $0.2 - 0.02 + 0.001333 = 0.1813$. Calculator: $\ln(1+\sin 0.2) = 0.1812$

Question 2

(a) $f(0) = \tan^{-1} 1 = \frac{\pi}{4}$. $f' = \dfrac{2}{1+(2x+1)^2} = \dfrac{1}{1+2x+2x^2}$, $f'(0) = 1$. $f'' = -\dfrac{2+4x}{(1+2x+2x^2)^2}$, so $f''(0) = -2$

(b) $f(x) \approx \dfrac{\pi}{4} + x - x^2$

Question 3

(a) $f' = (1-x^2)^{-1/2}$, $f'' = x(1-x^2)^{-3/2}$. $f(0) = 0$, $f'(0) = 1$, $f''(0) = 0$, $f^{(3)}(0) = 1$. So $\sin^{-1} x \approx x + \dfrac{x^3}{6}$

(b) $\displaystyle\int_0^{1/2} \left(x + \tfrac{x^3}{6}\right) \mathrm{d}x = \left[\tfrac{x^2}{2} + \tfrac{x^4}{24}\right]_0^{1/2} = 0.125 + 0.002604 = 0.127604$

(c) By parts: $\int \sin^{-1} x \,\mathrm{d}x = x \sin^{-1} x + \sqrt{1-x^2}$. Between 0 and $\frac{1}{2}$: $\dfrac{\pi}{12} + \dfrac{\sqrt{3}}{2} - 1$ $(= 0.12783)$

Question 4

(a) $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots$. Squaring: $x^2 - x^3 + \left(\frac{1}{4} + \frac{2}{3}\right)x^4 = x^2 - x^3 + \frac{11}{12}x^4$

(b) $x^2 - x^3 + \frac{11}{12}x^4 = 2x^3$ gives $x^2\left(\frac{11}{12}x^2 - 3x + 1\right) = 0$. $x = \dfrac{3 - \sqrt{9 - 11/3}}{11/6} = 0.38$ (taking the smaller root, which is within the range of validity of the series)

Question 5

(a) $e^x = 1 + x + \dfrac{x^2}{2} + \dfrac{x^3}{6} + \ldots$ and every term after $1+x$ is positive for $x > 0$, so $e^x > 1 + x$

(b) Let $x = t - 1$ with $t > 1$ (so $x > 0$): $e^{t-1} > t$, so $e^t > et$

(c) Take $t = \dfrac{4}{e}$ (> 1 since $e < 4$): $e^{4/e} > e \cdot \dfrac{4}{e} = 4$. Raising both sides to the power e: $e^4 > 4^e$. So e^4 is greater